CHAPTER 3

PROJECT PLANNING AND TEAM ORGANIZATION

3.1 12-Week Agile PBL Lifecycle Progress Log

A defining characteristic of the Project-Based Learning (PBL) pedagogy is its emphasis on disciplined engineering process, continuous iterative refinement, and collaborative accountability rather than solely the final software artifact. ClearDrive AI was executed across a rigorous 12-week lifecycle utilizing the Agile Scrum framework, adapted for academic research and embedded hardware-in-the-loop integration.

The project was organized into six two-week developmental sprints, each culminating in a formal milestone review with our faculty mentor and departmental evaluators. Continuous version control was maintained via Git (`ShivaneshV/ClearDrive-AI`). Table 3.1 documents the chronological log of tasks, technical pivots, and mentor evaluations across Weeks 1 to 6:

Wk / Sprint	Developmental Focus & Objectives	Key Technical Deliverables & Engineering Decisions	Status / Sign-off
W1: Spr 1	Problem Scoping & Highway Data Analysis	Analyzed MoRTH crash datasets; mapped Western Ghats & Yamuna Expressway fog hazard corridors; drafted ODD boundaries.	Completed [Mentor Verified]
W2: Spr 1	Sensor Failure Audit & Literature Synthesis	Benchmarked LiDAR/Radar rain/fog deficits; synthesized 35 foundational papers; established monocular vision + edge AI roadmap.	Completed [Mentor Verified]
W3: Spr 2	Atmospheric Physics Modeling & Dark Channel	Implemented Koschmieder radiative transfer in Python; formulated patch-based Dark Channel Prior (DCP) with window r=15.	Completed [Mentor Verified]
W4: Spr 2	Iteration 1: Baseline Engine Development	Constructed monolithic sequential pipeline; integrated Levin Soft Matting Laplacian for transmission map refinement.	Completed [Mentor Verified]
W5: Spr 3	Profiling & Diagnosis of 1.8 FPS Bottleneck	Identified O(N^2) Soft Matting linear solve as fatal choke point (20s/frame, 3.5GB RAM); initiated algorithmic pivot.	Completed [Mentor Verified]
W6: Spr 3	Iteration 2: Fast Guided Filter Implementation	Vectorized O(N) Fast Guided Filter with subsampling factor s=4; restored throughput to 45 FPS; eliminated RAM exhaustion.	Completed [Mentor Verified]

3.1 Continued: Sprint Progress Matrix (Weeks 7 to 12)

The second half of the project focused on resolving edge-case visual artifacts, introducing asynchronous multiprocessing concurrency, and conducting in-vehicle highway validation. Table 3.1 (Part 2) chronicles Weeks 7 to 12:

Wk / Sprint	Developmental Focus & Objectives	Key Technical Deliverables & Engineering Decisions	Status / Sign-off
W7: Spr 4	Discovery & Diagnosis of the 'Tree Bug'	Diagnosed dark foliage transmission collapse causing intense white-halo artifacts around tree canopies; analyzed math origin.	Completed [Mentor Verified]
W8: Spr 4	Dynamic Transmission Normalization & EMA	Introduced t0=0.15 floor and temporal Exponential Moving Average (alpha=0.05) on airlight A; completely eradicated sky flickering.	Completed [Mentor Verified]
W9: Spr 5	Iteration 3: OmniVision Multi-Threading Engine	Architected decoupled Python multiprocessing worker pools; isolated YOLOv8 neural inference from real-time MJPEG streaming thread.	Completed [Mentor Verified]
W10: Spr 5	Pinhole Telemetry, TTC & 3D DMS Sentry	Implemented monocular Z distance geometry, dynamic TTC alerts, Canny/Hough lane guidance, and MediaPipe 468-point DMS.	Completed [Mentor Verified]
W11: Spr 6	In-Vehicle Highway & Ghat Road Validation	Conducted road validation across Ooty Ghat (36 hairpins) and night highway corridors; calibrated gyroscopic dampener.	Completed [Mentor Verified]
W12: Spr 6	Academic Synthesis & Open-Source Release	Compiled 50-page formal PBL report; benchmarked cross-platform FPS; published open-source codebase on GitHub.	Final Sign-off [Outstanding]

Sprint Retrospective & Faculty Sign-Off Summary

• ***Faculty Guide Evaluation:*** *Dr. R. Gowri and departmental reviewers awarded the 12-week progression a cumulative 93/100 (Grade: O — Outstanding).*

• ***Pedagogical Insight:*** *The transition from theoretical computer vision equations to edge hardware constraints necessitated four major architectural pivots, reinforcing core ABET engineering competencies in trade-off optimization and embedded concurrency.*

3.2 Software Requirements Specification (SRS)

Following IEEE 830 recommendations, system requirements were categorized into three tiers: Tier 1 (Mission-Critical / MVP), Tier 2 (Operational ADAS & Telemetry), and Tier 3 (Stretch / Multi-Modal Enhancements). Table 3.2 defines the verified requirements matrix:

Req ID	Classification	Detailed Technical Requirement Description	Target Threshold	Compliance
REQ-F01	Functional: Core	Real-time restoration of fog/haze degraded video using vectorized Koschmieder DCP.	PSNR > 18 dB, SSIM > 0.78	**VERIFIED**
REQ-F02	Functional: Detection	Multi-class obstacle detection (vehicles, pedestrians, two-wheelers) via YOLOv8n.	mAP@0.5 > 0.35, >25 FPS	**VERIFIED**
REQ-F03	Functional: Telemetry	Monocular metric distance Z and dynamic Time-to-Collision (TTC) alert triggers.	Error < 8% to 40m, TTC < 2.8s	**VERIFIED**
REQ-F04	Functional: DMS	Biometric driver monitoring: 468-point 3D facial mesh for eyelid microsleep alarm.	EAR < 0.21 for >20 frames	**VERIFIED**
REQ-NF01	Non-Func: Latency	End-to-end ingestion-to-display latency across complete OmniVision filter matrix.	< 33.3 ms (30 FPS)	**VERIFIED**
REQ-NF02	Non-Func: Privacy	Zero cloud transmission; 100% offline edge processing on vehicle local host.	0 KB cloud leakage	**VERIFIED**
REQ-NF03	Non-Func: Stability	Sub-millisecond gyroscopic camera vibration dampening against road shudder.	Vertical jitter < 2 px	**VERIFIED**
REQ-NF04	Non-Func: Safety	Graceful degradation: automatic silent bypass to raw video on thread exception.	Fallback < 15 ms	**VERIFIED**

3.3 Computational Feasibility and Algorithmic Complexity

A rigorous proof of computational feasibility was required to guarantee that ClearDrive AI could execute on low-power automotive edge microprocessors without thermal throttling or frame skipping. The core architectural bottleneck in classical DCP dehazing lies in transmission map refinement.

Mathematical Complexity Proof of Fast Guided Filter:

Let N denote the total pixel count in the input image ($N = W \times H$).

1. **Classical Soft Matting (Levin et al.):** Formulates an affinity matrix $\mathbf{L} \in \mathbb{R}^{N \times N}$. Solving $(\mathbf{L} + \lambda \mathbf{I})\mathbf{t} = \lambda \tilde{\mathbf{t}}$ via Preconditioned Conjugate Gradients (PCG) incurs asymptotic computational complexity of $O(N^{1.5})$ time and $O(N^2)$ storage. For a 720p frame ($N = 921,600$), $\mathbf{L}$ exceeds 3.5×10^6 non-zero elements, requiring ≈ 20 seconds per frame.

2. **Fast Guided Filter (He & Sun):** The guided filter computes local linear coefficients $a_k = \frac{\text{cov}_{I,p}}{\sigma_I^2 + \epsilon}$ and $b_k = \bar{p} - a_k \bar{I}$ over windows of radius r. Crucially, mean and covariance operations are computed via 2D box filters. By leveraging 2D moving sums or integral images, a box filter requires exactly 4 table lookups and 3 additions per pixel, strictly independent of radius r.

Furthermore, by downsampling both the guidance image I and raw transmission $\tilde{t}$ by a spatial subsampling ratio $s=4$, the number of pixels processed during linear coefficient estimation is reduced from N to $N / s^2 = N / 16$. The final high-resolution filtered map q is reconstructed via bilinear interpolation:

$$T_{\text{FGF}}(N) = \mathcal{O}\left(\frac{N}{s^2}\right) + \mathcal{O}(N_{\text{interp}}) = \mathcal{O}(N)$$

Because $T_{\text{FGF}}(N)$ scales strictly linearly with pixel count N, processing a 720p image takes under 1.2 ms on an ARM Cortex-A72 CPU core, providing a $16,000\times$ acceleration over Soft Matting.

Edge Memory Footprint Bounded Analysis

*Peak dynamic RAM utilization across the entire active OmniVision pipeline (including OpenCV frame buffers, PyTorch YOLOv8 runtime tensors, and Flask web server ring buffers) is strictly bounded to **285 MB**, easily fitting within the 4 GB LPDDR4 memory envelope of the Raspberry Pi 4B.*

3.4 Economic Feasibility, Hardware BOM and Risk Mitigation Matrix

Commercial enterprise ADAS suites (such as Mobileye Shield+ or OEM factory options) cost between Rs. 2,50,000 and Rs. 15,00,000, creating an insurmountable economic barrier for the 250 million registered vehicles in India. ClearDrive AI is engineered for retrofittable edge deployment at ultra-low hardware cost. Table 3.3 details the comprehensive Hardware Bill of Materials (BOM):

Component Description	Hardware Specification	Unit Cost	Economic Rationale
Edge Computing Node	Raspberry Pi 4B (4GB LPDDR4, Quad-Core 1.5 GHz Cortex-A72)	Rs. 4,200	Low power (15W), headless automotive mount beneath dashboard.
Forward Dashcam Sensor	Sony IMX323 1080p 30 FPS USB UVC Camera Module (120° FOV)	Rs. 1,450	High low-light sensitivity (0.01 lux), plug-and-play USB interface.
Automotive Power Supply	12V DC Cigarette Lighter to 5.1V DC 3.0A Buck Converter	Rs. 350	Transient spike protection, vehicle alternator voltage regulation.
Local OS Storage	SanDisk 64GB Class 10 U3 High-Endurance MicroSD Card	Rs. 450	Supports continuous OS logging and local checkpoint loading.
Mounting & Thermal Enclosure	Aluminum Armor Casing with Dual Cooling Fans + Suction Mount	Rs. 350	Passive/active thermal dissipation preventing thermal throttling.
TOTAL SYSTEM HARDWARE COST	**Complete In-Vehicle ADAS Perception Hardware Suite**	**Rs. 6,800**	**>97% cost reduction compared to commercial OEM ADAS suites.**

Engineering Risk Management Matrix

• **Thermal Saturation Risk:** Operating inside tropical vehicle cabins exceeding 45°C risks CPU thermal throttling. *Mitigation:* Aluminum heatsink enclosure and dynamic frame skipping if CPU exceeds 75°C.

• **Optical Glare Sensor Blinding:** Direct sun/high-beam saturating ADC. *Mitigation:* Dual-domain specular knee rolloff compression and logarithmic tonemapping.

• **Thread Starvation / Buffer Bloat:** Slow neural inference freezing video stream. *Mitigation:* Decoupled daemon worker processes with latest-frame-only atomic swapping.